**CMSC 27100-3** Due: No due date

# Problem Set 9.5

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## 1 Recurrence Relations

**Problem 1:** The recurrence relation for the Towers of Hanoi problem (`https://en.wikipedia.org/wiki/Tower_of_Hanoi`) is:

$$H_n = 2H_{n-1} + 1$$

where $H_1 = 1$. Find a closed form for $H_n$ via the guess-and-check method.

**Problem 2:** Solve the following recurrence relation using the geometric series formula:

$$a_n = 2(a_(n-1)) + 7$$

where $a1 = 8$.

**Problem 3:** Solve the following homogeneous recurrence relation:

$$a_n = 3a_(n-1) + 10a_(n-2)$$

for $a_1 = 1$ and $a_2 = 33$.

**Problem 4:** Use your previous solution to solve the inhomogeneous recurrence relation

$$a_n = 3a_(n-1) + 10a_(n-2) + 12$$

for $a_1 = 2$ and $a_2 = 21$.